

MAX YUAN

647-819-0765 • maxyuan081205@gmail.com • Markham, Ontario

EDUCATION

University of Waterloo | Mechatronics Engineering, Honours Co-op | Sept 2026 - Sept 2031

TECHNICAL SKILLS

Languages: Java, Python, C++

Skills: Robotics Simulation, Reinforcement Learning (PPO), PID control, Odometry, Inverse Kinematics

Libraries/Tools: Isaac Sim, Isaac Lab, rsl_rl, ROS2, Rviz, Onshape, OpenCV, YOLOv8, VLA

Strengths: Proactive learner, Self-driven, Passionate, Feedback-driven

PROJECTS

The Bracket Way — Autonomous VLA Sandwich Assembly on Bracket Bot | Sept 12 – Sept 13 2026

1st Place @ BOTS Hackathon (WAT.AI VS UTMIST, Battle Of The Schools)

- Fine tuned $\pi 0$ VLA foundation model via supervised-fine tuning collecting 100+ teleoperated input and output pairs (video footage + prompt as input, joint position + velocity as outputs)
- Preprocessed training data with grid resampling, aligning mismatched 65 Hz joint and 5 fps camera streams onto a shared fixed time grid to eliminate frame-action mislabeling
- Prevented model overfitting by integrating data aggregation, introducing randomization in object positioning and lighting
- Built a finite state machine utilizing both classical (IK + CV) and learning-based (VLA) models in each respective state.
- Detected & localized lettuce with HSV threshold segmentation, utilizing inverse kinematics for manipulation & accuracy, adding wrist-camera closed-loop correction for 1–2 cm head-camera calibration error

Custom Quadruped Locomotion — Unitree Go2 PPO in Isaac Lab | Aug 2026 – Sept 2026

Solo project: custom ManagerBasedRLEnv for flat-terrain Go2 locomotion, trained with PPO via NVIDIA Isaac Lab

- Custom built RL env: 48-dim observation space, 12-DOF joint-position actions, and custom reward stack
- Implemented a sigmoid curriculum that ramps the foot-lift reward forcing the robot to master balance before refining its gait
- Repo: github.com/NotMax08/unitree_go2_ppo_isaacLab

Autonomous Maze Navigation — Nova Carter in Isaac Sim | May 2026 – July 2026

Solo project: full ROS2 Nav2 stack with a Nova Carter differential-drive robot via NVIDIA Isaac Sim 5.1 on Ubuntu

- Built end-to-end autonomous navigation: SLAM mapping with slam_toolbox, AMCL localization, Nav2 planning with Regulated Pure Pursuit controller, ROS2 as a middleware framework and Rviz for debugging
- Constructed the full TF tree (map → odom → chassis_link → sensors) from scratch via Isaac Sim action graphs; debugged QoS mismatches, sensor rate bottlenecks, transformation frame authority conflicts
- Diagnosed inconsistent warped SLAM map through topic loop rates caused by scan misalignment during turns
- Repo: github.com/NotMax08/isaac-sim-maze-slam-nav2

Co-Captain & Lead Programmer — FIRST Tech Challenge Robotics Team #18844 | Sept 2023 – Apr 2025

2025 FIRST World Championships — 1st Connect Award Winner / Ochoa Division

- Point-to-Point PID movement system: tuned gains for 3 positional PID controllers (x, y, heading) enabling autonomous pathing with minimal error
- Implemented 2-wheel odometry + IMU for robot localization, converting local pose into global coordinates
- Automated sample pickup system with real-time object detection using OpenCV and a custom-trained CNN for distance and sample orientation

LEADERSHIP EXPERIENCE

Co-Captain & Outreach Coordinator — FIRST Tech Challenge Robotics Team #18844 | Sept 2023 – Apr 2025

- Co-led Team 18844 PrOteens as Team Canada representatives on an international stage, overseeing programming systems integration and technical decisions